

Chat server

```
#include<mpi.h>
#include<stdio.h>
int main(int argc,char *argv[])
{
    int a[2],c,e,f,d,r;
    MPI_Init(&argc,&argv);
    MPI_Status s;
    MPI_Comm_rank(MPI_COMM_WORLD,&r);

    if(r==0)
    {
        printf("\n Enter two values: ");
        scanf("%d %d",&a[0],&a[1]);
        MPI_Bcast(&a,2,MPI_INT,0,MPI_COMM_WORLD);
    }

    if(r==1)
    {
        MPI_Bcast(&a,2,MPI_INT,0,MPI_COMM_WORLD);
        c=a[0]+a[1];
        printf("sum=%d\n",c);
    }
    if(r==2)
    {
        MPI_Bcast(&a,2,MPI_INT,0,MPI_COMM_WORLD);
        d=a[0]-a[1];
        printf("Difference=%d\n",d);
    }

    if(r==3)
    {
        MPI_Bcast(&a,2,MPI_INT,0,MPI_COMM_WORLD);
        e=a[0]*a[1];
        printf("Product=%d\n",e);
    }

    if(r==4)
    {
        MPI_Bcast(&a,2,MPI_INT,0,MPI_COMM_WORLD);
        f=a[0]/a[1];
        printf("Division=%d\n",f);
    }

    MPI_Finalize();
    return 0;
}
```

Output

```
mpicc chat.c -o ch  
mpirun -np 5 ch
```

```
Enter two values: 10 5  
Division=2  
sum=15  
Difference=5  
Product=50
```